

Name: Longhai LiDate: June 30, 2026College: Arts & ScienceDepartment: Mathematics & Statistics

**INFORMATION FOR UPDATE OF CURRICULUM VITAE – NEW ITEMS
(ITEMS NOT PREVIOUSLY REPORTED)**

The cut off date for items to be reported is JUNE 30.

1. PERSONAL

- Official webpage: <https://artsandscience.usask.ca/profile/LLi>
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Department of Mathematics & Statistics
University of Saskatchewan
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Saskatoon, SK, S7W0G8 CANADA

2. DEGREES

- Ph.D., University of Toronto, 2007, Statistics
Supervisor: Radford M. Neal
Thesis: Bayesian Classification and Regression with High Dimensional Features
- M.Sc., University of Toronto, 2003, Statistics
- B.Sc., University of Science and Technology of China, 2002, Statistics.

4. Employment History

- Full Professor, July 1st, 2018, Dept. of Math & Stat., Univ. of Saskatchewan, SK, Canada
- Associate Professor, July 1st, 2012, Dept. of Math & Stat., Univ. of Saskatchewan, SK, Canada
- Assistant Professor, July 1st, 2007, Dept. of Math & Stat., Univ. of Saskatchewan, SK, Canada
- Research Intern, Nov. 2006 to Feb. 2007, Microsoft Research, Redmond, WA, USA
- Sessional Instructor, 2006-2007, University of Toronto, Toronto, ON, Canada

9. TEACHING ACTIVITIES**9.1 Scheduled Instructional Activity****2025-2026**

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 851	T2	Linear Statistical Models	LEC	4	39	156
STAT 443	T2	Linear Models	LEC	3	39	117
STAT 442	T2	Statistical Inference	LEC	3	39	117

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 850	T2	Mathematical Statistics and Inference	LEC	1	39	39
STAT 845	T1	Statistical Methods for Research	LEC	10	39	390
MATH 996	T1T2	Research Supervision (Ph.D.)	RES	1		
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		

2024-2025

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 348	T1	Sampling Techniques	LEC	15	39	585
STAT 812	T1	Computational Statistics	LEC	5	39	195
STAT 420	T1	Topics in Computational Statistics	LEC	1	39	39
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		
BIOS 996	T1T2	Research Supervision (M.Sc.)	RES	1		
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		

9.4 Teaching Materials

2025-2026

- [4] STAT 442/850: a textbook entitled “Theory of Statistical Inference” in PDF (≥ 200 pages) and HTML.
- [3] STAT 845: a textbook entitled “Statistical Inference and Learning Methods for Research” in PDF (≥ 200 pages) and HTML
- [2] STAT 443/851: a textbook entitled “Theory of Linear Model” in PDF (≥ 200 pages) and HTML.

2024-2025

- [1] STAT 443/851: 273 pages of handwritten lecture notes were developed and posted to students.

9.5 Other Teaching-Related Activities

2025-2026

- [1] Peer evaluation for Matthew Schmirler, Mar. 2026.

10. SUPERVISION AND ADVISORY ACTIVITIES

10.1 Undergraduate Student Supervision

2025-2026

- [2] George Chen, B.Sc., Computer Science, Simon Fraser University, June 1, 2025 – Aug. 31, 2025, Supervised.
- [1] Shruti Kaur, B.Sc., Computer Science and Statistics, May – Aug., 2025, Supervised.

10.2 Graduate Student Supervision

- [2] Jing Wang, Ph.D., Biostatistics, School of Public Health, Co-supervised with Prof. Li Xing, 2023–2026 (Transferred to other supervisor)
- [1] Dananji Egodage, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Cindy Feng, 2023–2025 (Defended: Aug. 30, 2025)

10.3 Graduate Theses Supervised

2025-2026

- [2] Dananji Egodage, 2025, Component-wise Z-residuals for Checking Bayesian Hurdle Models, M.Sc., defended on Aug. 30, 2025.

2024-2025

- [1] Effie Wuqian Gao, 2024, Z-residuals for Checking Bayesian Hurdle Models, M.Sc., defended on Aug. 30, 2024.

10.4 Supervision of Post-Doctoral Fellows and Research Associates

- [2] Tingxuan Wu, University of Saskatchewan, Postdoc Research Associate, Jan. 2025 – Dec. 2026.
- [1] Ming Ming Zhang, MITACS Postdoc, 2022-2025.

10.6 Thesis Committee Memberships

Table 1: Thesis Committee Memberships

#	NAME	DEG	PRO-GRAM	TIME FRAME	ROLE
31	Tian-sui Wu	M.Sc.	Biosta-tistics	2025–Pre-sent	Chair
30	Prab-hawi Kahat-apitiye	Ph.D.	Statis-tics	2025–Pre-sent	Mem-ber
29	Rasel Kabir	Ph.D.	Biosta-tistics	2025–Pre-sent	Chair
28	Mo-ham-mad Toran-jsimin	M.Sc.	Biosta-tistics	2023–2025 (Def. 09/2025)	Mem-ber
27	Lina Li	Ph.D.	Biosta-tistics	2023–Pre-sent	Mem-ber
25	Mangladeep Bhullar	Ph.D.	Physics	2023–2025 (Def. 11/2025)	Cog-nate
22	Han Wang	Ph.D.	Sociol-ogy	2020–Pre-sent	Cog-nate

12. PAPERS IN REFERRED JOURNALS

2025-2026

- [6] Wu, T., Gao, WE, Feng, C., and Li, L., Z-residuals for Diagnosing Bayesian Models, *Journal of American Statistical Association*, under revision.
- [5] Wu, T., Li, L., and Feng, C., Z-residuals Diagnostics for Cox Proportional Hazards Models: Distinguishing Functional Form Misspecification from Nonproportional Hazards, with an Application to Biliary Cirrhosis Survival Times, *Canadian Journal of Statistics*, Accepted on June 5, 2026.
- [4] Nolan, J., Su, C., Li, L., 2025. Evaluating Railroad Duopoly Behavior: A Market Level Analysis. *Review of Network Economics* 24, 87–111. <https://doi.org/10.1515/rne-2025-0034>

2024-2025

- [3] Wu, T., Feng, C., Li, L., 2025. Cross-validatory Z-Residual for Diagnosing Shared Frailty Models. *The American Statistician*, 79(2), 198–211. <https://doi.org/10.1080/00031305.2024.2421370>
- [2] Wu, T., Li, L., Feng, C., 2025. Z-residual diagnostic tool for assessing covariate functional form in shared frailty models. *Journal of Applied Statistics*, 52(1), 28–58. <https://doi.org/10.1080/02664763.2024.2355551>
- [1] Wu, T., Feng, C., Li, L., 2024+. A Comparison of Estimation Methods for Shared Gamma Frailty Models. *Statistics in Biosciences*, accepted 24 June 2024. <https://doi.org/10.1007/s12561-024-09444-7>

14. PRESENTATIONS

14.1 Invited Presentations

2025-2026

- [4] Z-residuals for Checking Bayesian Models. Presented at: University of Calgary, Calgary, AB, Canada; July 28, 2025
- [3] Sparse Learning for Assessing the Association Between Gut Microbiome and Parkinson’s Disease. Presented at: The 3rd JCSDS, Hangzhou, China, July 13, 2025.

2024-2025

- [2] Z-residuals for Checking Bayesian Models. Presented at: International Conference on Statistics and Data Science, Vancouver, BC, Canada; June 24, 2025
- [1] Z-residuals for Checking Bayesian Hurdle Models. Presented at: EcoStat 2024; July 17, 2024; Beijing, China.

15. REPORTS AND OTHER OUTPUTS

15.1 Software Released Publicly

- [4] Wu, T. and Li, L., 2026. **Zresidual**: Computing and Diagnosing Gaussian-like Residuals. [Github] [Demo]. Version 0.1-0 on Github (2026).
- [3] Li, L., 2026. R Functions for Computing Z-residuals for **survreg** and **coxph** Objects. [URL].
- [2] Li, L. and Liu, S., 2019–2026. **HTLR**: Bayesian Logistic Regression with Hyper-LASSO priors. DOI: 10.32614/CRAN.package.HTLR. [CRAN] [Github] [URL]. Version 0.4 (2019), version 0.4-1 (2019), version 0.4-2 (2020), version 0.4-3 (2020), version 0.4-4 (2022), version 1.0 (2026).
- [1] Li, L., 2011–2026. **BCBCSF**: Bias-corrected Bayesian Classification with Selected Features. DOI: 10.32614/CRAN.package.BCBCSF. [CRAN] [URL]. Version 0.0-0 (2011), version 0.0-1 (2011), version 0.0-2 (2012), version 1.0-0 (2013), version 1.0-1 (2015), updated to version 1.0-2 (2026).

17. RESEARCH FUNDING HISTORY

2025-2026

- [2] **NSERC Individual Discovery Grant (No. 2026-07053)** – *Prediction-based Methods for Statistical Learning and Inference in Biosciences and Epidemiology*, \$185,000 (37K per year), 2026-2031, PI.
- [1] **CANSSI Collaborative Research Team Projects** – *Statistical Methodologies and Computational Tools to Identify Microbial Correlates of Canadian Bee Gut Health*, Project 29, 2025-2028, Co-PI.

18. PRACTICE OF PROFESSIONAL SKILLS

2025-2026

- [14] Refereeing for *Journal of Statistical Computation and Simulation*, June, 2026
- [13] Refereeing for *Journal of Statistical Computation and Simulation*, April, 2026
- [12] Refereeing for *Journal of the Royal Statistical Society: Series C*, April 2026
- [11] Refereeing for *Bioinformatics*, March 2026
- [10] Refereeing for *Journal of Computational and Graphical Statistics*, March 2026
- [9] External Referee for a tenure and promotion of SFU, Dec. 2025
- [8] External Examiner for the doctoral thesis by Xiaoqing Zhang at University of Regina, Dec. 8, 2025
- [7] Refereeing for *Journal of Statistical Computation and Simulation*, Dec. 2025
- [6] Refereeing for *Journal of Computational and Graphical Statistics*, Sept. 2025
- [5] External Examiner for the doctoral thesis by Na Zhang at University of Alberta, August 28, 2025
- [4] Refereeing for *Journal of the Royal Statistical Society: Series C*, August 2025
- [3] Refereeing for *Journal of Applied Statistics*, August 2025

2024-2025

- [2] Organizing an invited Session for the 7th Symposium of ICSA Canada Chapter, McGill University, August 2026
- [1] Organizing an invited Session for 2025 SSC Annual Meeting, Saskatoon, SK, Canada, June 2025

19. ADMINISTRATIVE SERVICE

19.1 University Committees

2025-2026

- [2] Chair, Collaborative Biostatistics Program, University of Saskatchewan.
- [1] USASK NSERC Discovery Grant (DG), Internal Reviewer.

19.2 College and Departmental Committees

2025-2026

- [5] Member, Budgeting and Planning Committee, Dept of Math & Stat
- [4] Co-Chair, Undergraduate Committee (Statistics), Dept of Math & Stat
- [3] Member, Graduate Program Committee in Statistics, Dept of Math & Stat

2024-2025

- [2] Member, Graduate Program Committee in Statistics, Dept of Math & Stat
- [1] Search subcommittee for a faculty position in statistics

20. PROFESSIONAL OR ASSOCIATION OFFICES AND COMMITTEE ACTIVITY

2024-2025

[2] Member of NSERC Discovery Grant EG 1508 Committee

[1] Local Organizing Committee, 2025 Annual Meeting of the Statistical Society of Canada held at the U of S

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Nothing to report for timeframe: 2025-2026